The Normal Distribution

Section 1: The Empirical Rule (68-95-99.7)

1 In a normal distribution with mean $\mu = 100$ and $\sigma = 15$, approximately what percentage of scores fall between 70 and 130?

a. 95%

b. 68%

c. 99.7%

d. 50%

2 Using the 68-95-99.7 rule, what percentage of observations in a normal distribution fall above $\mu + 1\sigma$?

a. 16%

b. 32%

c. 34%

d. 2.5%

3 If heights of adult men are $N(70, 3)$ inches, what percentage are taller than 76 inches?

a. 2.5%

b. 5%

c. 16%

d. 0.15%

Section 2: Finding Normal Probabilities

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4 For a standard normal variable Z, find $P(Z < -1.28)$ given that the standard normal CDF at -1.28 is approximately 0.10.

a. 0.10

b. 0.90

c. 0.20

d. 0.05

5 If $P(Z < 1.96) \approx 0.975$, what is $P(-1.96 < Z < 1.96)$?

a. 0.95

b. 0.975

c. 0.05

d. 0.025

6 A machine fills cereal boxes normally with $\mu = 16.2$ oz and $\sigma = 0.1$ oz. What proportion of boxes contain less than the stated label of 16.0 oz?

a. $P(Z < -2.0) \approx 0.0228$

b. $P(Z < -1.0) \approx 0.1587$

c. $P(Z > 2.0) \approx 0.9772$

d. 0.5000

Section 3: Inverse Normal Calculations (Finding Values from Probabilities)

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7 Test scores are normally distributed with $\mu = 500$ and $\sigma = 100$. What score corresponds to the 90th percentile ($z \approx 1.28$)?

a. 628

b. 590

c. 600

d. 696

8 In a normal distribution, what are the z-scores corresponding to the first quartile (Q_1) and third quartile (Q_3)?

a. $z \approx -0.67$ and $z \approx +0.67$

b. $z \approx -1.00$ and $z \approx +1.00$

c. $z \approx -0.50$ and $z \approx +0.50$

d. $z \approx -1.96$ and $z \approx +1.96$

9 The weights of bags of potato chips are $N(\mu, 0.4 \text{ oz})$. If the heaviest 5% of bags weigh more than 16.66 oz ($z_{0.95} = 1.645$), find the mean weight μ.

a. 16.0 oz

b. 17.3 oz

c. 15.8 oz

d. 16.2 oz

Section 4: Assessing Normality with Normal Probability Plots

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10 What appearance in a Normal Probability (Q-Q) Plot indicates that the sample data comes from a roughly normal population?

a. The plotted points follow a roughly linear pattern

b. The plotted points form a bell-shaped curve

c. The plotted points are completely horizontal

d. The plotted points form a curved parabolic shape

11 If a Normal Probability Plot shows a clear upward curve (concave up), what does this indicate about the distribution?

a. The distribution is skewed to the right

b. The distribution is skewed to the left

c. The distribution is perfectly symmetric

d. The distribution has no spread

12 What is the total area under any valid probability density curve?

a. 1

b. 100

c. 0.5

d. Depends on the standard deviation

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